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Pedestrian Tracking Using YOLO and Multi-Object Tracking Algorithms

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Abstract

This thesis deals with the implementation of a system for pedestrian tracking using YOLO detection and the multi-object tracking algorithms ByteTrack and DeepSORT. The system combines the YOLOv8 detection model with tracking algorithms for tracking multiple pedestrians in real time. Three YOLO models were trained on different datasets and tested in combination with both tracking algorithms. The results are evaluated on the MOT17 dataset using standard MOT metrics. The program allows switching between detection models and tracking algorithms.

Keywords: YOLO — ByteTrack — DeepSORT — Multi-Object Tracking — MOT Challenge — Pedestrian Tracking

Supplementary Material: [Demonstration video](#) — [Downloadable code](#) — [Models and Outputs](#)

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1. Introduction

Real-time pedestrian tracking is an important task in computer vision with applications in security systems, autonomous vehicles, and human motion analysis. The goal of this thesis is to implement a system for multi-object pedestrian tracking that combines modern detection models with tracking algorithms.

Task definition: The task is to create a system capable of detecting and tracking multiple pedestrians in video sequences while assigning consistent IDs to individual objects throughout the entire video. The system must be able to handle occlusions, disappearances and reappearances of objects, as well as dynamic scenes with varying numbers of pedestrians.

2. Object Detection Using YOLO

YOLO (*You Only Look Once*) is a modern approach to object detection that enables fast and accurate real-time detection. The main idea of YOLO is that all bounding box predictions and object class predictions are generated in a single pass of the image through a convolutional neural network, which significantly reduces computational complexity compared to classical

methods based on *region proposals*, such as R-CNN or Fast R-CNN [3].

During processing, the image is divided into a grid (for example $S \times S$). Each grid cell is responsible for predicting objects whose center lies within that cell. For each cell, several bounding boxes are predicted, where each box contains the center coordinates (x, y) , width w , height h , and a confidence score (*objectness*) that expresses the probability that an object is present in the box. In addition, class probabilities are predicted for each box, which are combined with the objectness score when computing the final prediction score.

The original version of YOLO (YOLOv1) predicted a fixed number of boxes per grid cell and did not include the concept of *anchor boxes*. Newer versions (YOLOv2 and above) introduced *anchor boxes*, which are predefined box shapes to which the network predicts offsets and scale, improving the detection of objects of various sizes and shapes. YOLOv8 returned to *anchor-free* detections. Modern YOLO versions, such as YOLOv5 and YOLOv8, use an architecture consisting of three parts:

- **Backbone:** a convolutional network that extracts

image features at multiple scales, typically a large CNN trained for object classification.

- **Neck:** a convolutional architecture that combines information from different scales and improves the detection of both small and large objects.
- **Head:** a convolutional network that predicts the final bounding boxes, confidence scores, and object classes on the output grids of each scale. During inference, *Non-Maximum Suppression* (NMS) is applied to remove overlapping boxes and retain only those with the highest confidence.

This approach allows YOLO to combine high detection speed with sufficient accuracy, making it suitable for real-time multi-pedestrian tracking in dynamic scenes [4].

3. Multi-Object Tracking Algorithms

Multi-Object Tracking (MOT) is the task of tracking multiple objects over time, where each object is assigned a unique ID. In this thesis, two modern tracking algorithms are implemented: ByteTrack and DeepSORT.

3.1 ByteTrack

ByteTrack [8] is a tracking algorithm based on associating every detection box, including those with low confidence scores. The key idea is that even weak detections can contain useful information for tracking. The algorithm uses a Kalman filter for position prediction and the IoU (Intersection over Union) metric for associating detections with existing tracks.

3.2 DeepSORT

DeepSORT [5] combines a Kalman filter with a deep learning model for extracting appearance features. It uses a ReID (Re-Identification) model to generate feature vectors from bounding boxes, enabling better object association even under occlusion. In this implementation, the OSNet model from the torchreid library is used, providing 512-dimensional feature vectors.

4. Datasets

For pedestrian detection, it is appropriate to use a dataset that contains images of people in various situations, poses, and environments. It is important that the dataset includes pedestrians of different sizes, frontal and side views, varying degrees of occlusion, and different pedestrian densities. Such data allow the model to learn a robust representation and correctly detect pedestrians in real-world scenarios.

4.1 CityPersons

CityPersons [6] is a popular pedestrian detection dataset based on the Cityscapes semantic segmentation dataset [1]. It contains approximately 5 thousand images from urban environments with pedestrian annotations, including occlusions and various body sizes. The dataset was collected using a camera mounted on a vehicle driving through the city, so the images are captured approximately at waist height.

4.2 WiderPerson

WiderPerson [7] is a large-scale dataset focused on pedestrian detection in various environments, including both indoor and outdoor scenes, with different densities and body sizes. Most images are captured outdoors, not only on streets but also at sports events, celebrations, or in mountainous areas. The dataset provides very high scene and situation diversity, enabling the training of models capable of detecting pedestrians under diverse conditions.

5. Experiments and Evaluation

This chapter describes experiments focused on training detection models, their mutual comparison, and subsequent integration with multi-object tracking algorithms. The goal of the experiments was to analyze the impact of different training datasets on pedestrian detection capability and to verify system behavior in both real-world and benchmark conditions.

5.1 Training of Detection Models

For all experiments, the YOLOv8 model in the *medium* configuration was used, implemented using the *Ultralytics* library. This architecture was chosen as a compromise between accuracy and computational complexity, making it suitable for real-time applications.

Three separate detection models were trained:

- a model trained exclusively on the CityPersons dataset,
- a model trained exclusively on the WiderPerson dataset,
- a model trained on a combination of the CityPersons and WiderPerson datasets.

The goal of this division was to examine how the choice of training data affects the model's ability to generalize to different types of scenes and environments.

5.2 Detection Evaluation

The trained models were evaluated on the test splits of the CityPersons and WiderPerson datasets. Each model

was tested on both datasets, allowing a comparison of its generalization capability outside the training domain.

The results show that the model trained on the WiderPerson dataset achieves better performance when tested on unseen data, which can be attributed to the high diversity of scenes and environments as well as the large number of images in this dataset. In contrast, the model trained exclusively on the CityPersons dataset shows better performance in urban scenes but weaker generalization outside this environment.

The best results on both test datasets were achieved by the model trained on the combination of CityPersons and WiderPerson datasets, confirming that combined training data lead to a more robust detection model. These conclusions are illustrated by a graphical comparison of detection metrics in Figure 1.

5.3 Integration with Tracking Algorithms

The detection models were subsequently integrated with two multi-object tracking algorithms: ByteTrack and DeepSORT. Both algorithms were implemented using custom implementations and adapted for use with YOLO detector outputs.

A qualitative comparison of both approaches was performed visually using comparative videos that demonstrate differences in ID stability, behavior under occlusions, and object re-identification 2.

In this particular video, the model trained on the CityPersons dataset detects the highest number of objects, which is likely due to the video being captured from a low camera position and depicting exclusively pedestrians at an intersection, closely matching the characteristics of this dataset.

5.4 Evaluation on MOT17

The complete system, consisting of a detection model and a tracking algorithm, was further evaluated on the MOT17 benchmark [2]. Each combination of detection model and tracking algorithm was tested separately, and the results were evaluated using standard MOT metrics.

The experiments show that detection quality has a significant impact on overall tracking performance. Models trained on combined datasets achieved more stable results and better metrics compared to models trained on a single dataset. At the same time, performance differences between the ByteTrack and DeepSORT algorithms could be observed 3.

Literature

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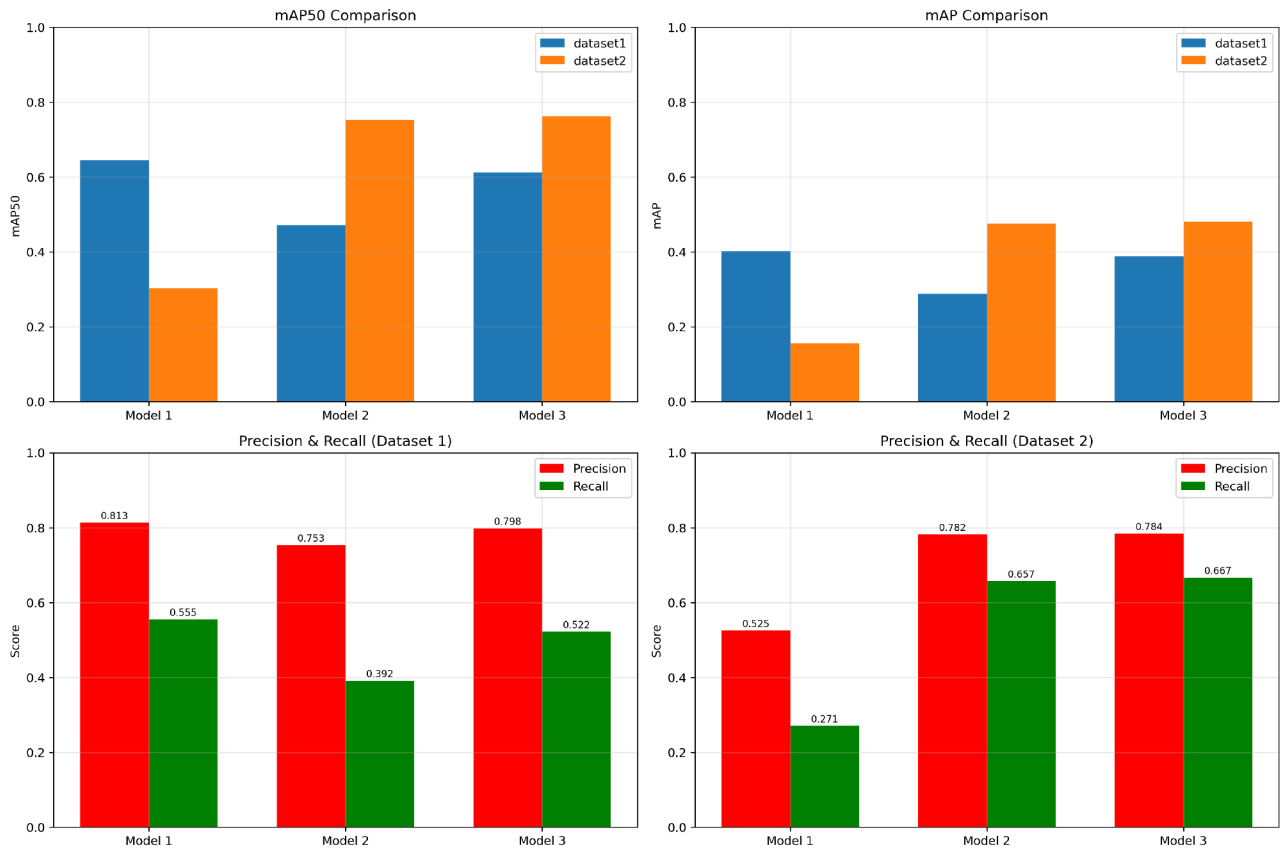


Figure 1. Comparison of detection metrics of trained YOLO models on the CityPersons and WiderPerson datasets. Model 1 is trained on CityPersons, Model 2 on WiderPerson, and Model 3 on the combination of both datasets.

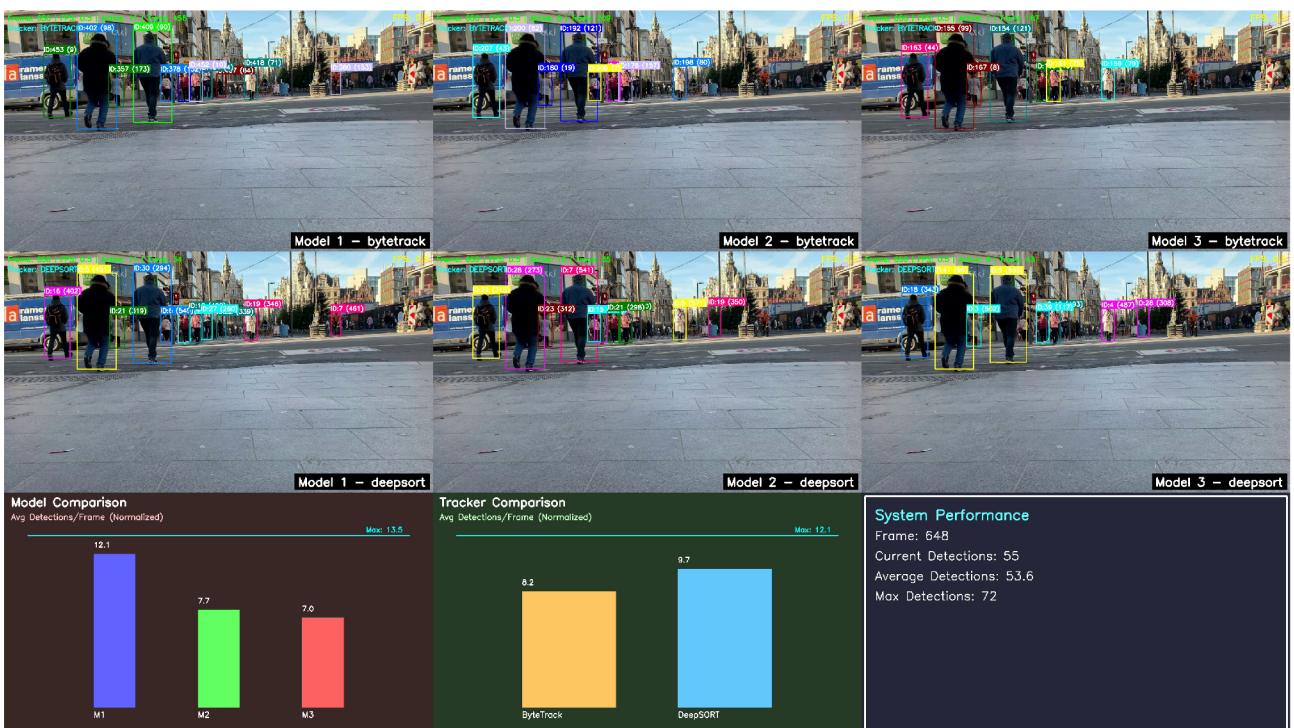


Figure 2. Visual comparison of the ByteTrack and DeepSORT tracking algorithms on the same video sequences.

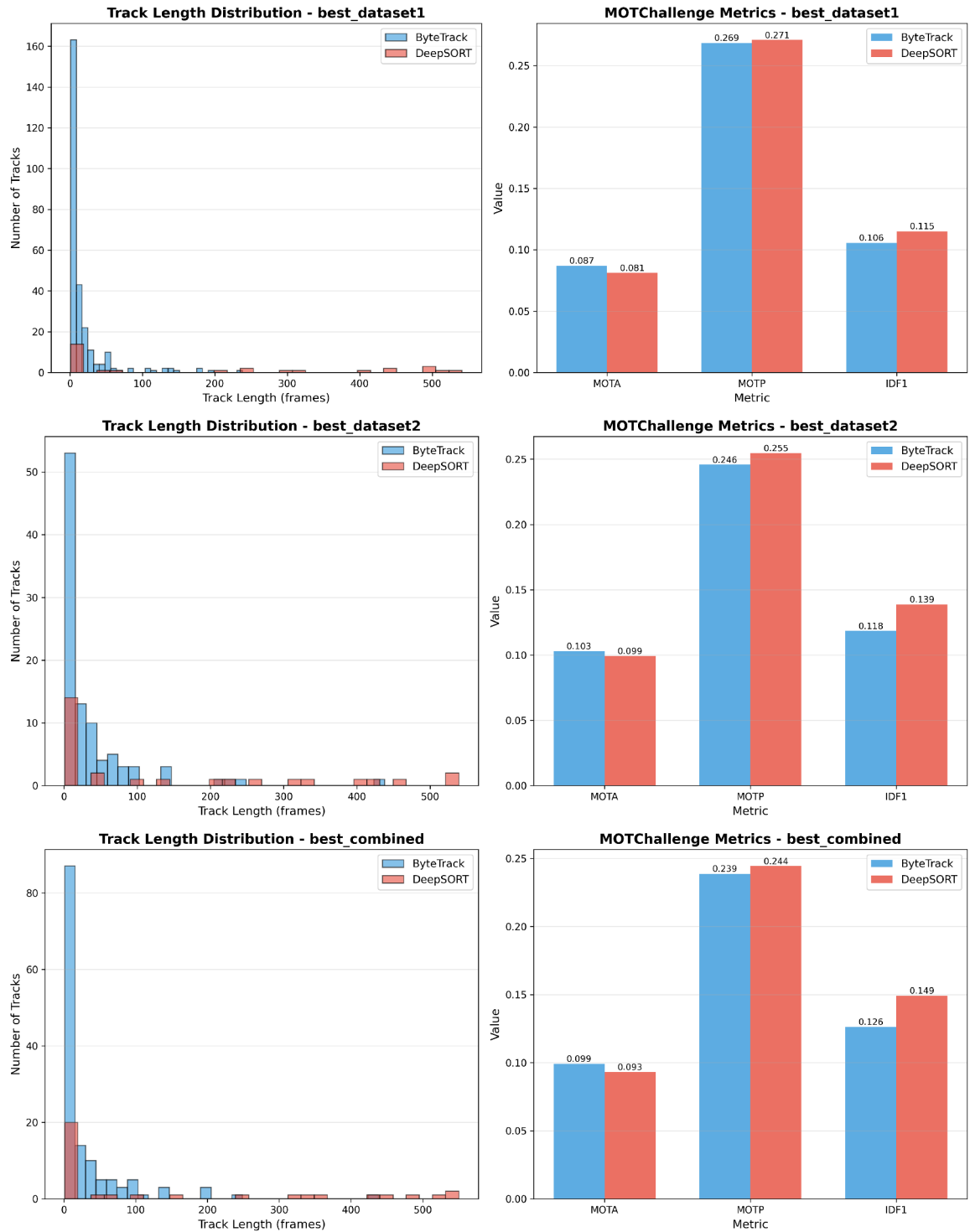


Figure 3. Evaluation of combinations of detection models and tracking algorithms on the MOT17 benchmark using standard MOT metrics. The graphs are shown for the model trained on the CityPersons dataset, the WiderPerson dataset, and the combination of both datasets, respectively.